Lesson plan · for the teacher

Good data, fair AI

AI · Level 1 · Lesson 3

Time
40 min

Materials
Board and chalk (or paper on the wall); Scrap paper and pencils; Two or three local fruits, or drawings of them; Worksheet: Spot the data gap (one per pair, optional)

Learning goals

- Explain that a machine that learns from examples can make more mistakes when some examples are missing.
- Spot who or what is missing from a set of examples.
- Suggest a fair way to fill a data gap, such as adding examples from the people left out.

Activity: Whose examples?: Test it

1. Warm-up (5 min): Ask: has a phone or app ever got someone's name or words wrong? Take 3–4 stories. Write "Why?" on the board.
2. Draw examples (8 min): Each pupil draws one fruit on scrap paper and names it. Put all the drawings in the "Our examples" box. Count each kind together.
3. Test it (10 min): The class is the recogniser and may only answer with a fruit in the box. Show a common fruit, then a local fruit nobody drew. Pupils vote for the closest drawing.
4. Pair talk (10 min): Pairs discuss: what was missing, and how could we fix it? Then ask: if voice typing learned mostly from some accents, whose voices might it get wrong?
5. Wrap-up (7 min): Add drawings of the missing fruit and test again. Write on the board: "It's not just how many examples — it's whose." Then do the exit ticket.

Draw a big box on the board labelled "Our examples". Bring two or three local fruits pupils may not think to draw, such as guava, dates, prickly pear, jackfruit or a baobab pod.

Discussion

- Which drawing on the board is closest to the new fruit?
- Is that the right answer?
- So how could we make our examples fairer?

Exit ticket

- One thing missing from our examples, and one way to fix it.

Try it in class tomorrow
Play "Whose examples?": Pupils draw one fruit each and stick the drawings on the board — these are the examples. The class is a "fruit recogniser" that can only answer with fruit on the board. Hold up a local fruit nobody drew. Ask: what went wrong, what was missing, and how is this like voice typing and accents?

Worksheet · for pupils

Good data, fair AI

AI · Level 1 · Lesson 3

Name _____ Class _____ Date _____

1 **Voice typing makes more mistakes for Salma than for her colleague. What is the most likely reason?**

- ◯ Salma doesn't speak clearly enough
- ◯ Voices like hers were rare in the recordings it learned from
- ◯ Someone set the phone up to be unfair to her
- ◯ Voice typing never works for anyone

2 **Computers can't be unfair because they're machines.**

◯ True ◯ False

3 **Which of these could cause a data gap?**

- ☐ Training photos taken mostly in one country
- ☐ Voice recordings from only a few accents
- ☐ Examples from the people who will use the tool
- ☐ A malaria tool trained mainly on one parasite type
- ☐ Testing the tool with many different groups

4 **Match each data gap to a fix.**

Voice typing trained on a few accents	• • • •	Add scams in Kiswahili and other languages
A malaria tool trained on one parasite	• • • •	Test with a balanced set of faces
Face tools tested mostly on lighter skin	• • • •	Add photos of the other parasite types
A scam filter trained only on English	• • • •	Add recordings of other accents

5 **Complete the sentence.**

An AI system is only as good as its _____. When some groups are _____ from them, it often makes more _____ for those groups.

Word bank: examples · missing · mistakes · rules · money

✎ **Think of an AI tool your pupils might meet. Who in your class might it work less well for — which languages, accents or faces?**